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From food waste to sustainable aviation fuel: cobalt molybdenum catalysis of pretreated hydrothermal liquefaction biocrude

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Valorization of wet biowaste into sustainable aviation fuel presents a promising opportunity to decarbonize the aviation industry. While hydrothermal liquefaction of food waste produces high energy content biocrude oil, its complex composition introduces challenges for achieving drop-in fuel properties. This research demonstrates a pathway from food waste-derived biocrude to sustainable aviation fuel through single-stage hydrotreating using cobalt molybdenum catalyst. Hydrotreating parameters are screened via the Taguchi method and optimized via Response Surface Methodology. The fuel candidate meets American Society for Testing and Materials jet fuel properties without blending, including low sulfur limits, density, viscosity, flash point, and freeze point. Further, Circularity Index, the fraction of renewable and recovered resources, demonstrates that compared to conventional jet fuel, this pathway improves energy and carbon circularity by 31.1% and 17.0%, respectively. Additionally, this approach identifies future direction, including reaction and catalyst tuning for increased isomerization, to improve fuel properties.

Rising energy demand and the negative impacts of greenhouse gas (GHG) emissions have hastened efforts to produce sustainable aviation fuel (SAF). To achieve the United States's SAF Grand Challenge of meeting 100% domestic demand for aviation fuel and net-zero carbon emissions, a significant uptake in SAF is required. SAF can form a circular carbon loop that utilizes carbon emissions during production, transportation, and combustion, effectively reducing aviation's carbon footprint[1]. Further, aviation represents 2–3% of global GHG emissions and 11% of U.S. GHG emissions[2,3]. However, SAF can reduce carbon emissions by 80% compared to traditional jet fuel[3–5], largely due to the use of biogenic waste material, which allows uptake of feedstock to offset combustion carbon emissions from SAF[3].

Currently, there are eight SAF production pathways and three co-processing pathways approved by the American Society for Testing and Materials (ASTM)[6]. Synthesized paraffinic kerosene from hydroprocessed esters and fatty acids (HEFA-SPK) is the most mature and least expensive pathway, but it is severely constrained by feedstock availability[7]. HEFA-SPK hydroprocesses lipid feedstocks, including FOGs, algae, vegetable oils like soybean oil, and seed oils like jatropha. Furthermore, synthesized paraffinic kerosene from hydroprocessed hydrocarbons, esters, and fatty acids (HC-HEFA-SPK) utilizes an additional feedstock, algae oil from *Botryococcus braunii*, to produce SAF. Fischer-Tropsch hydroprocessed synthesized paraffinic kerosene (FT-SPK) converts woody biomass (e.g., municipal solid waste, agricultural

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waste, forest/wood residues, energy crops) into syngas through gasification, followed by production of hydrocarbons through FT, and hydroprocessing. The additional production of aromatics (FT-SPK/A) addresses the low aromatics content of FT-SPK fuel through the alkylation of light aromatics with FT-derived olefins. Synthesized isoparaffins from hydroprocessed fermented sugars (HFS-SIP) utilizes microbial conversion of sugars from cellulosic biomass (e.g., corn stover, non-woody biomass, pretreated FOGs), to produce farnesene, which can then be hydroprocessed into isoparaffins. Alcohol-to-jet (ATJ) synthetic paraffinic kerosene produces isobutanol or ethanol via fermentation or gasification of cellulosic biomass, which then undergo dehydration, oligomerization, and hydrogenation to form jet-range hydrocarbons. ATJ synthetic paraffinic kerosene with aromatics (ATJ-SKA) also addresses the low aromatic content of ATJ SAF, deriving aromatics from C2-C5 alcohols through additional dehydration and aromatization steps. Lastly, hydrothermal conversion of fatty acid esters and fatty acids (CHJ) treats waste and energy oils through catalytic hydrothermal liquefaction and hydroprocessing. The blend limit is 50% for all approved pathways, except for HFS-SIP and HC-HEFA SPK, which are limited to 10%. For co-processing, esters and fatty acids, FT hydrocarbons, and HEFA hydrocarbons are hydroprocessed with petroleum in a conventional refinery, limited to 5%, 5%, and 10% blends, respectively.

The utilization of waste biomass plays a significant role in the production of biofuels for the transportation sector and reduction of GHG emissions. Biofuels can be defined as first-generation (edible oil seeds), second-generation (non-edible oil seeds, waste materials), third-generation (algae), and fourth-generation (genetically engineered oil crops) according to their feedstock[8]. In particular, second-generation feedstocks avoid land use and enable the recovery of carbon and energy from waste streams, presenting a sustainable feedstock for fuel production. Approximately 11% of bio-based energy and chemicals in the U.S. are produced from waste biomass[9]. Potential biomass feedstocks from waste streams include waste lipids like FOGs (fat, oil, and grease), food waste, algal blooms, sewage sludge, and agricultural and forestry residues. While catalytic hydroprocessing of lipid feedstocks, such as waste cooking oils, is currently the most competitive pathway for renewable diesel and SAF, feedstock sustainability is limited by availability and cost[10]. One promising feedstock for SAF is food waste, an impactful biomass stream that represents reusable carbon sources for renewable energy applications. In the United States, food waste is the most common discarded material for landfill and incineration[11]. Over 30% of food is wasted or lost in the global supply chain each year; In 2021, it was reported that 13.2% was lost in the chain from farm to consumer, while an additional 17% was wasted at the consumer level, including households, food service, and retail[12]. Furthermore, disposal of food waste in landfills has further environmental impacts through GHG emissions during decomposition[11]. Food waste is an ideal feedstock for hydrothermal liquefaction (HTL), a thermochemical conversion method that reacts wet biomass under high heat and pressure to produce biocrude oil, along with aqueous, char, and gaseous byproducts. Compared to other thermochemical technologies, HTL has an economic advantage by using the inherent water content of feedstocks as a reaction medium, thereby avoiding the time-consuming and cost-prohibitive drying pretreatment step[13]. Moreso, it can convert a broad range of feedstock, including non-lipid components of biomass, such as proteins and carbohydrates[14]. While HTL biocrude is a promising substitution for liquid petroleum products such as transportation fuels, it possesses several undesirable physicochemical properties that make it unsuitable as a drop-in fuel, including high oxygen content; high viscosity and acidity; and low heating value[13,15]. Furthermore, biocrude oil derived from biogenic feedstocks contain alkali impurities and sulfur that can cause catalyst deactivation over time[16]. Nonetheless, HTL's ability to convert a broad range of feedstocks gives it an advantage over other SAF conversion technologies, such as HEFA-SPK, which is sensitive to feedstock volatility, price, and availability[7]. For example, the minimum fuel selling price (MFSP) of HEFA-SPK in the United States ranged from $1.84–9.40/GGE, depending on feedstock source[7]. Comparatively, the MFSP of SAF from HTL has been reported from $2.77/GGE for sewage sludge feedstock[17], $3.34/GGE for food waste[18], and up to $5.42/GGE for cultivated algae[19]. Furthermore, 8-18% of costs were associated with the required upgrading of HTL biocrude oil, including hydrotreating and hydrocracking[17,19]. For feedstocks requiring deeper heteroatom removal, the addition of a hydrodenitrogenation unit only added $0.02/GGE[19].

Valorization of biowaste feedstocks like food waste into renewable transportation fuels such as SAF promotes both feedstock and energy sustainability, as well as a zero-waste circular bioeconomy. Furthermore, this waste-to-energy paradigm contributes to the United Nations Sustainable Development Goals (SDG) 7, 12, and 13. In the 2023 SDG Report, it was found that although 28.2% of electricity came from renewable energy sources, growth in clean energy for the transportation sector has been much more limited, with only 4.0% from renewable sources[12]. The aviation industry, like most of the transportation sector, is difficult to electrify, so HTL represents a potential solution to pivot away from fossil fuels and achieve SDG 7 Affordable and Clean Energy. Furthermore, SAF from food waste directly serves SDG 12 Responsible Consumption and Production by providing a renewable alternative to alleviate fossil fuel consumption, as well as utilizing a feedstock that helps the goal to decrease food waste and loss 50% by 2030[12]. Lastly, SAF production and consumption improves carbon circularity, therefore working toward SDG 13 Climate Action to reduce GHG emissions and mitigate global warming[12].

HTL biocrude must be upgraded to remove inorganic components and improve its fuel properties. Depending on the feedstock, different challenges and upgrading routes exist. For example, HTL feedstocks with high protein content, such as algae, typically produce nitrogenous biocrude that may require deeper heteroatom removal with more severe or multi-stage hydrotreating[20,21]. Lignocellulosic oil contains a significant amount of aromatics desirable for SAF, but the complex composition of lignin compounds make them recalcitrant to hydrotreating, necessitating reaction engineering and catalyst development for efficient hydrodeoxygenation[22]. Furthermore, HTL of high-lipid feedstocks typically produces heavier biocrude compounds due to repolymerization reactions, requiring hydrocracking during upgrading[23]. Common technologies used for biocrude upgrading include distillation, esterification, emulsification, hydrotreating, hydrocracking, and steam reforming[15,24,25]. Catalytic hydrotreating can target removal of heteroatoms, as well as improve hydrogen content and fuel quality of the biocrude, while avoiding carbon loss from decarbonylation and decarboxylation reactions[20]. Previous studies have explored hydrotreating biocrude from various feedstocks including lignocellulose, agricultural waste, algae, sewage sludge, waste cooking oil, and food waste. Molybdenum-based catalysts are desirable for their low cost, comparable activity to noble-metal catalysts, and effectiveness toward deoxygenation (de-O). Cobalt molybdenum catalysts have been shown to be slightly more effective for denitrogenation (de-N) compared to nickel molybdenum. For example, 99% de-N was achieved for hydrotreated HTL biocrude using CoMo[26], while NiMo achieved up to 80% using single-stage hydrotreating[27], and 92.5% using two-stage hydrotreating[21]. Additionally, Cronin et al. found that CoMo led to increased removal of heterocyclic nitrogenates, such as pyridine and pyrimidine[28]. Catalysts with higher metal loads exhibit improved yield and de-O[29], and while organic solvents such as methanol, ethanol, and 2-propanol can act as hydrogen donors, they may also cause competing reactions between hydrogenating the solvent and biocrude oil[30].

Hydrotreating temperatures for HTL biocrude typically range from 350 to 400 °C and pressures of 4–10 MPa[20,27,28]. The use of lower

temperatures (200-325 °C) has also been explored to improve biocrude stabilization for increased demetallization[31,32] and de-N[21]. Depending on the reactor configuration (batch vs. continuous), the catalyst load and retention time (RT) may be defined individually or as a weight hourly space velocity (WHSV, h^{-1}). In batch reactors, the equivalent WHSV can be defined as the ratio of biocrude to catalyst mass divided by the RT. For continuous hydrotreating, the 0.3–1.0 h^{-1} WHSV was used, with 0.5 h^{-1} the most common[27,28,31,32]. Meanwhile, batch reactors applied catalyst:oil mass ratios of 0.5 and RTs of 4 h, yielding an equivalent WHSV of 0.5 h^{-1}, making them comparable to conditions in a continuous reactor[20,21,32]. Existing studies showed that temperature and pressure are significant factors for de-O and de-N, while also influencing fuel quality and product yields[29,33]. Although higher temperature increased oil recovery and is required for further de-N, it may also lead to polymerization, condensation, and cracking reactions due to instability of reactive oxygenates[32,34]. At higher hydrogen pressure, the extent of de-O increased, while coking and decarboxylation reactions were minimized and there was improved hydrogenation of heterocyclic nitrogenates[20,27,30]. However, at severe pressures, polymerization of small molecules may also occur, decreasing the final oil yield[29]. Meanwhile, low pressures may result in the diffusion of gases formed during hydrocracking reactions, causing biocrude oil to be increasingly converted into gas and lighter molecules[29]. The RT also influenced product yields and fuel quality, with longer times allowing for further de-O and hydrocracking reactions[29,30]. But at extended reaction times (>4 h), polymerization reactions occured and increased coke formation caused catalyst deactivation[29,30].

Although previous works have also produced SAF candidates through the catalytic treatment of food waste-derived precursors, their resulting fuel viability was restricted by blending limits, nitrogen, and sulfur levels. For example, Cronin et al. hydrotreated biocrude from HTL of food waste using CoMo and NiMo, but 0.86–1.34 wt% nitrogen and 0.02–0.03 wt% sulfur remained[28]. Meanwhile, catalytic upgrading of volatile fatty acids (VFA) from the anaerobic digestion of food waste produced a neat SAF candidate, i.e., unblended, with 26–68 ppm nitrogen, slightly above the ASTM limit of 15 ppm[16]. Additionally, the VFA-SAF candidate was limited by its flash point and low temperature viscosity, requiring at least 70% blending with Jet A to meet ASTM specification limits in Tier α and β tests[16]. In this study, it was shown that cobalt molybdenum catalyst effectively upgraded food waste-derived HTL biocrude oil into drop-in SAF. Through parameter screening and optimization, a neat SAF candidate was produced that met preliminary ASTM qualifications for jet fuel properties, possessing low nitrogen and sulfur levels, thereby advancing circularity of the aviation sector.

Results and discussion

Hydrotreating parameter screening

The products yields, elemental composition, and parameter screening results are displayed in Fig. 1. It was seen that oil yield generally decreased with combinations of higher parameter levels (Fig. 1A). This was mainly due to polymerization reactions occurring during extended RTs that increased char formation, as well as increased hydrocracking and gasification reactions at high pressure and temperature. Greater amounts of aqueous and char products were seen at the low and medium temperatures, indicating that at higher severity, less polymerization and recondensation occurs over gasification. Lack of aqueous product at higher temperatures suggested decarboxylation and decarbonylation pathways over hydrodeoxygenation, yielding gaseous products rather than water. In general, higher de-O resulted in increased higher heating values (HHV) (Fig. 1A, B). 76–100% de-O was achieved, with up to 2.21 wt% remaining oxygen in the oil. Hydrotreated oil with remaining oxygen were treated at lower temperature and RT, indicating the importance of these parameters for de-O. Desulfurization (de-S) was effective at higher severities, with up to 94% sulfur removal and 20 ppm remaining. De-N was least efficient, ranging from 7 to 23% with 0.55–0.66 wt% nitrogen remaining in the hydrotreated oil.

Improved fuel quality and heteroatom removal were obtained at the expense of carbon and energy recovery (C_r and E_r), especially due to carbon losses from gasification at severe hydrotreating conditions. The most significant parameters affecting de-O, E_r, and C_r were determined using Taguchi delta values, the difference between the highest and lowest mean of response (Fig. 1C). For de-O, the most significant parameters were catalyst-to-oil ratio (cat:oil) > temperature (T) > hydrogen load-to-biocrude oxygen ratio (H:O) > retention time (RT), while for C_r and E_r they were T > cat:oil > RT > H:O. Though H:O was more important for oxygen removal than carbon and energy recovery, the highest average de-Os were obtained at 15:1 H:O ratios. Therefore, T, cat:oil, and RT were selected as the parameters to undergo optimization for SAF production while the H:O ratio was held constant.

Improved boiling point distribution and hydrocarbon composition was also observed across changes in hydrotreating temperature and catalyst ratio. Temperature was the most significant factor affecting the boiling point distribution of the hydrotreated biocrude oil (Fig. S1A). At lower temperatures, light fractions <250 °C increased compared to PT biocrude oil, but the amount of mid-range fractions from 250 to 350 °C decreased while the amount of heavier fuel fractions >350 °C increased. Therefore, at mild hydrotreating conditions, polymerization and coking reactions may decrease the yield of viable distillates in the transportation fuel range. Notably, at the highest

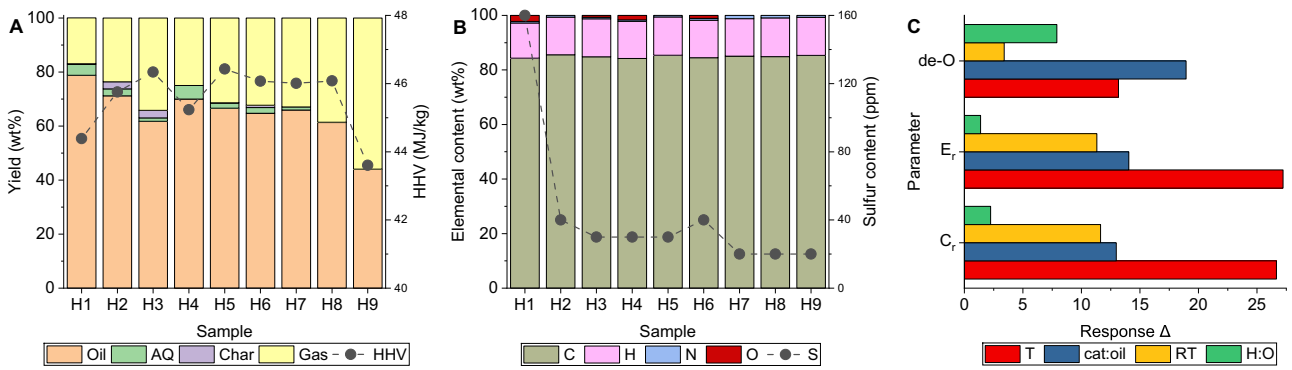


Fig. 1 | Taguchi parameter screening results for yields, oil quality, and parameter significance. A Mass balance of oil, aqueous (AQ), char, gas yields and higher heating value (HHV); **B** stacked bar elemental composition of carbon (C), hydrogen (H), nitrogen (N), oxygen (O), and sulfur (S); and **C** bar chart for Taguchi delta of response for the effect of temperature (T), catalyst:oil ratio (cat:oil), retention time (RT), and hydrogen:oxygen ratio (H:O) on deoxygenation (de-O), energy recovery (E_r), and carbon recovery (C_r). Source data are provided as a Source Data file.

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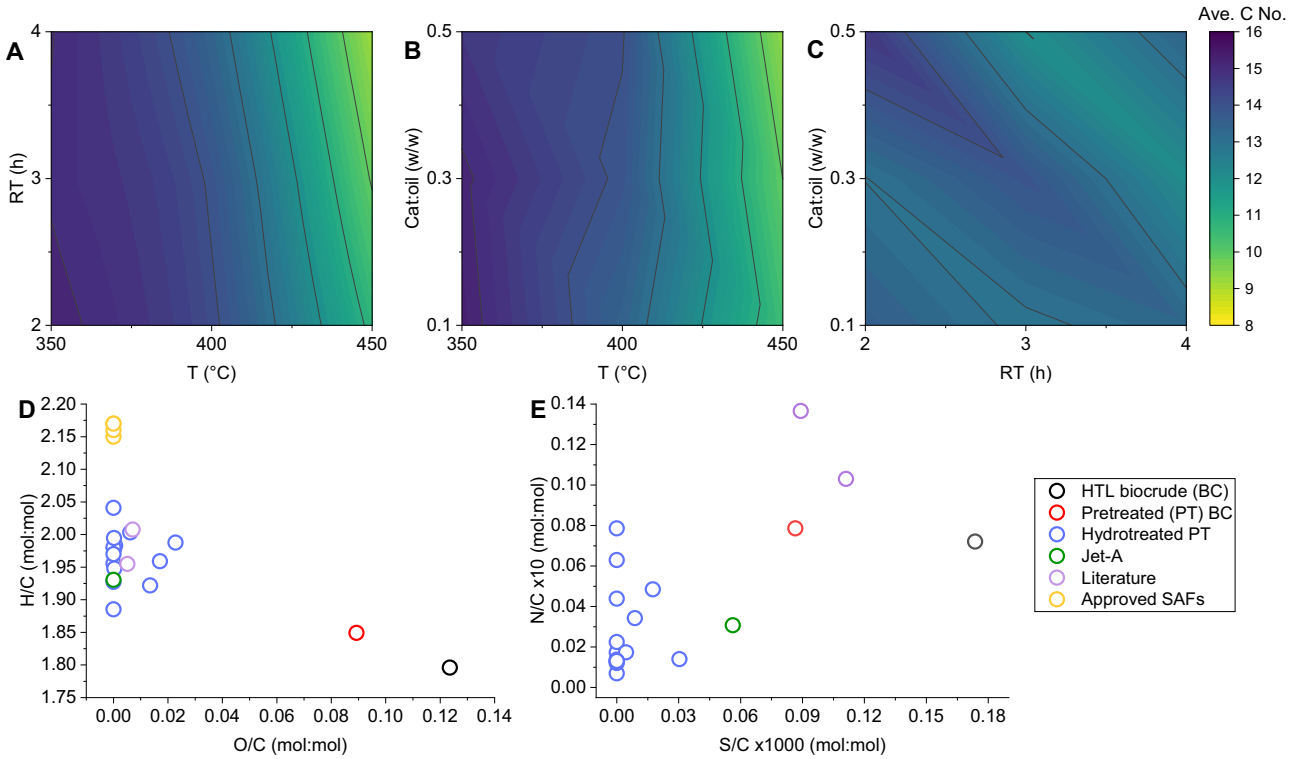


Fig. 2 | Hydrotreating parameter effects on average carbon number (Ave. C No.) and the resulting heteroatom contents. Heat map contour plot showing Ave. C No. (weighted average of the carbon number of chemical compounds) that make up the hydrotreated hydrothermal liquefaction (HTL) biocrude for **A** retention time (RT) vs. temperature (T); **B** catalyst:oil ratio (cat:oil) vs. temperature (T); **C** catalyst:oil ratio (cat:oil) vs. retention time (RT); and modified Van Krevelen diagrams comparing the elemental molar ratio of **D** H/C (hydrogen/carbon) vs. O/C (oxygen/carbon); **E** N/C x10 (nitrogen/carbon) vs. S/C x1000 (sulfur/carbon) of HTL biocrude (BC), pretreated (PT) BC, Hydrotreated PT from this work, commercial Jet-A, hydrotreated HTL biocrude from food waste in literature[28], and approved SAFs[16]. Source data are provided as a Source Data file.

temperatures, the hydrotreated biocrude oil contained >99% distillate fractions <250 °C, with H9 exhibiting extremely similar boiling point distribution to Jet-A. At the mildest conditions (H1), the amount of n-alkanes significantly increased from 19.91% in the PT biocrude oil to 84.72% after hydrotreating, indicating de-O and hydrogenation of unsaturated hydrocarbons and O-containing compounds such as fatty acids (Fig. S1B). With increased catalyst ratio (H2-3), alkenes were further hydrogenated, and small increases in aromatics and cycloalkanes were seen. At higher temperatures, isomerization and cracking reactions are favored, while n-alkanes can also undergo dehydrocyclization to form aromatics[35,36]. This caused the hydrocarbon distribution to further broaden and include more cycloalkanes, isoalkanes, and aromatics. At elevated temperature, higher oil yield and de-O also improved the C_r and E_r of the hydrotreated HTL biocrude (Fig. S1C).

Hydrotreating parameter optimization and fuel properties

Average carbon number and hydrocarbon type are significant properties in predicting the properties of SAF[37,38]. Having an average carbon number closer to conventional jet fuel's is desirable to stay within ASTM specification limits for SAF. Contour heat maps elucidated the effect of T, RT, and cat:oil on the average carbon number of hydrotreated biocrude oil (Fig. 2A–C and Table S5). It was seen that T had a greater effect than RT on average carbon number, with decreased carbon number correlated with increased hydrotreating T (Fig. 2A). The slight tilt of the contour lines toward the left also reflected decreased carbon number at longer RT, indicating breakdown of the carbon chains with limited repolymerization. Similar trends were seen for T and cat:oil, with decreased carbon number with increased T at all levels of cat:oil (Fig. 2B). Meanwhile, the change in carbon number across RT and cat:oil was less pronounced (Fig. 2C). A diagonal of slightly higher average carbon numbers was observed from high cat:oil and low RT to low cat:oil and high RT, showing that at mild combinations of cat:oil and RT, less cracking occurred. The diagonal pattern across the axis ranges was mainly attributed to the greater effect of T on average carbon number. For example, points on the contour line where average carbon number ~13, were those where different T (350 vs. 450 °C) were used at the same RT and cat:oil. Therefore, the change in T caused deviations in the average carbon number. Compared to T, RT, and cat:oil had less effect on carbon number. Overall, the average carbon number further decreased at combinations of higher T, RT, and cat:oil. Therefore, while a higher temperature was required to achieve an average carbon number closer to jet fuel, consideration should be taken in its combination with RT and cat:oil to avoid SAF products with carbon chains that are too light.

The modified Van Krevelen diagrams exhibit effective de-O and de-S during hydrotreating, but varying de-N (Fig. 2D–E). A majority of the SAF samples yielded complete oxygen and sulfur removal, with O/C and S/C molar ratios of 0. The H/C molar ratio of Jet-A and other conventional jet fuels depend on their specific composition. In this case, the obtained Jet-A sample was measured to have an H/C ratio of 1.93. Nearly all the hydrotreated HTL biocrude had H/C > 1.90 and some had H/C > 2.00, reflecting their comparable elemental composition to Jet-A, high HHV, and potential for improved combustion. A wide range of de-N was observed, indicating the importance of hydrotreating parameter tuning on nitrogen removal. Considering the cold-weld CHN method used, the variation may also be due to systematic (instrumental or measurement) errors. With a nitrogen detection limit of 0.1%, at least 87% de-N was achieved in a few cases. High de-N in a single hydrotreating step may be a result of the HTL

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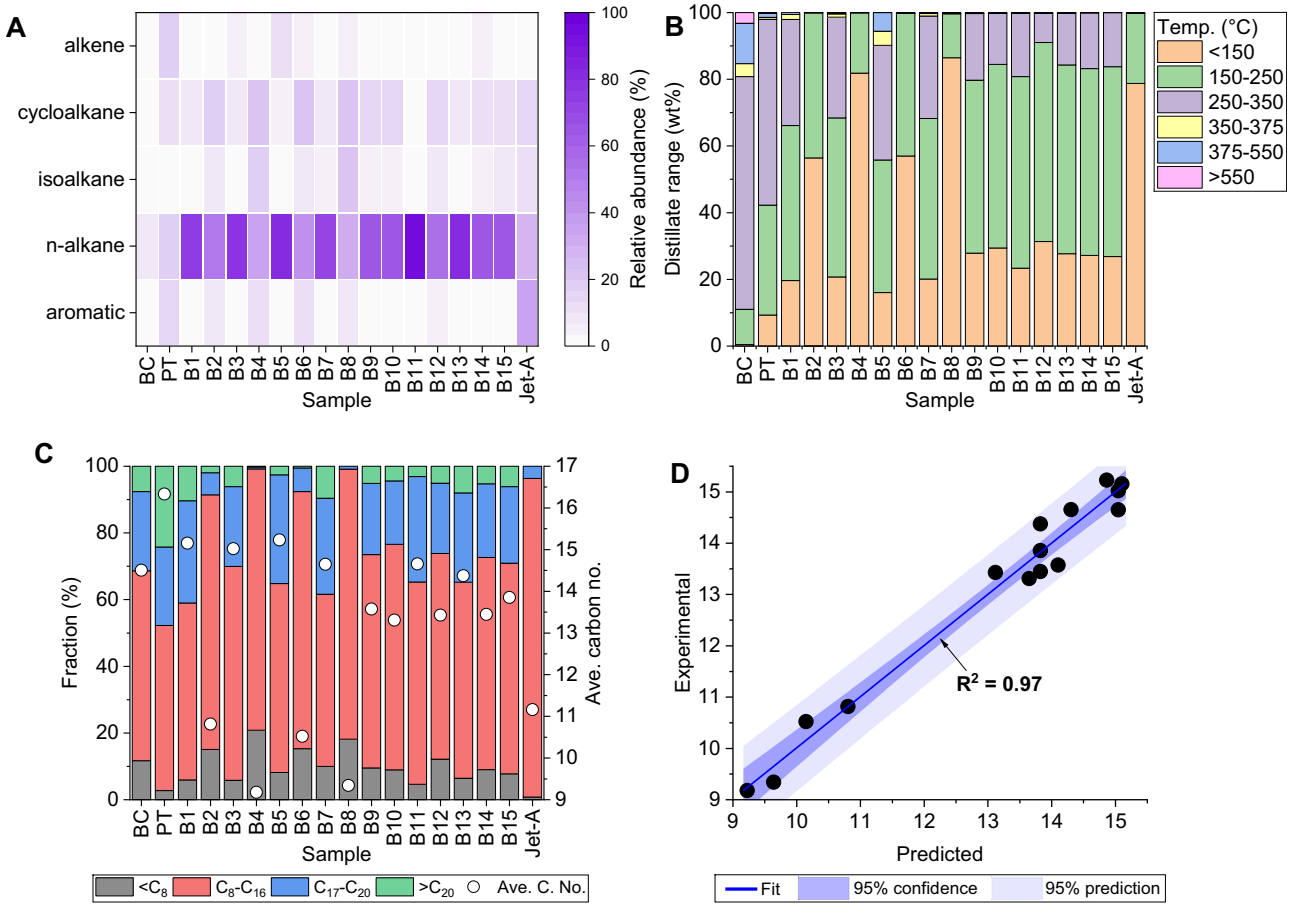


Fig. 3 | Physicochemical fuel properties of the hydrotreated HTL biocrude oil. **A** Heat map of the relative abundance (%) of aromatic, n-alkane, isoalkane, cycloalkane, and alkene content; **B** distillate fractions at different temperature ranges; **C** carbon number range fractions (%) and average carbon number; and **D** Experimental vs. predicted average carbon numbers from regression ($n = 15$ independent experiments with 2 replicates) of the hydrotreating data, where the center line is the linear fit ($y = 0.9978x + 0.032$, $R^2 = 0.97$, $p = 9.37 \times 10^{-5}$), dark and light blue shaded regions represent the 95% confidence and prediction band of the regression, respectively. Source data are provided as a Source Data file.

biocrude pretreatment, where the removal of salt and ash prevents catalyst plugging and deactivation[39]. Furthermore, up to 40% oxygen, 55% sulfur, and 22% nitrogen could be removed during pretreatment, enabling additional stabilization of the biocrude prior to hydrotreating[39]. Notably, compared to a previous study that used CoMo and NiMo catalysts to hydrotreat HTL biocrude oil derived from food waste, improved de-O, de-S, and de-N was achieved in this work[28]. However, in comparison to reported hydrocarbon composition of approved SAFs from FT-, HEFA-, and ATJ-SPK, the H/C value from HTL still has room to improve[16].

Comparing the hydrocarbon type, carbon number distribution, and boiling point distribution of HTL biocrude oil before and after hydrotreating, it was confirmed that hydrotreating at high temperature produced SAF samples with composition closest to Jet-A (Fig. 3). Hydrotreating successfully improved the hydrocarbon composition, indicated by an increase in the n-alkane content of all samples (Fig. 3A). Like the trend seen in the parameter screening tests, hydrotreating at combinations of higher temperature, RT, and catalyst load further improved the hydrocarbon type to contain increased amounts of aromatics, isoalkanes, and cycloalkanes. Notably, several of the SAF samples contained similar or higher amounts of isoalkanes and cycloalkanes compared to Jet-A, along with lower aromatic content. This could lead to improved cold flow properties and decreased contrails[40]. Notably, the Jet-A sample contained 35% aromatics and synthesized jet fuel must have 8-25% aromatics content. All samples hydrotreated at 450 °C (B2, B4, B6, B8) met the low level aromatics requirement of at least 8%, ranging from 9 to 13%. As seen in our SAF candidate, this value is also expected to increase after taking the jet cut from the whole hydrotreated biocrude due to the higher abundance of aromatics in the C8–C16 range. Recent work has also indicated that cycloalkanes can replace the volume swell properties of aromatics in jet fuel[41].

HTL biocrude oil consistently became lighter after hydrotreating. This was reflected in the boiling point distribution, where hydrotreating increased the fraction of distillate ranges <250 °C (Fig. 3B). Like what was seen during the parameter screening, hydrotreating at the lowest parameter levels (e.g., B5) caused polymerization reactions that formed heavier compounds with boiling points above 350 °C. On the other hand, hydrotreating at higher temperatures and combinations of increased RT and cat:oil produced samples mainly composed of fractions with distillate ranges <250 °C. Specifically, all samples hydrotreated at the highest temperature level had a boiling point distribution most similar to Jet-A.

The carbon number range in Jet-A is typically C8–C16. This was confirmed in the commercial Jet-A sample tested, where 95% of its compounds were within the C8-C16 range and its average carbon number was 11.2 (Fig. 3C). Chemical compounds were also grouped into <C8, C17–C20, and >C20, to represent potential distillates in the ranges for gasoline, diesel, and heavy oils, respectively. All samples that had an average carbon number similar to or less than Jet-A (<11.5) were those produced at high temperature during hydrotreating (450 °C). However, since they still contained fractions outside the

C8–C16 range, their average carbon number will likely change after distillation to obtain the SAF cut. Due to the importance of average carbon number on the chemical and handling properties of jet fuel, regression of the RSM data was used to predict the average carbon number and compare it to the experimental values (Fig. 3D). The predicted and experimental values were highly correlated, with an R^2 of 0.97. A 95% confidence band and prediction band also demonstrated the ability of the model to predict new values within the experimental parameters.

SAF candidacy: Tier α tests for hydrotreated HTL biocrude oil

Based on the parameter screening and optimization, the HTL biocrude was hydrotreated to obtain a SAF candidate for Tier α prescreening of jet fuel properties that are critical for combustor operability limit. The SAF candidate achieved a carbon and energy recovery of 85.2 and 90.2%, respectively. Complete de-O and de-N was achieved, along with a final sulfur content of 10 ppm, meeting the 15 ppm maximum[42]. The physicochemical properties were further listed in Table S6. Compared to conventional jet fuel, the SAF candidate from this work contained higher amounts of light hydrocarbons, was within ASTM specification limits for critical bulk properties, and had a comparable distillation range predicted within specification limits (Fig. 4). The hydrocarbon compositions of both the hydrotreated HTL biocrude oil and SAF cut were mainly n-alkanes, followed by cycloalkanes, aromatics, and iso-alkanes (Fig. 4A, B). Compared to the average jet fuel carbon distribution, the hydrotreated HTL biocrude oil contained high amounts of lighter (<C9) and heavier (>C13) hydrocarbons (Fig. 4A). The greater amount of heavy compounds in the whole hydrotreated HTL biocrude oil caused the average carbon number to be 12.7, slightly higher than jet fuel's 11.4. However, after taking the SAF cut from C9 to C13, the average carbon number was more similar to jet fuel, decreasing to 10.7 (Fig. 4B). This SAF cut represented 31.5 wt% of the hydrotreated HTL biocrude oil. Meanwhile, the light and heavy ends are also potential candidates for biobased gasoline and diesel.

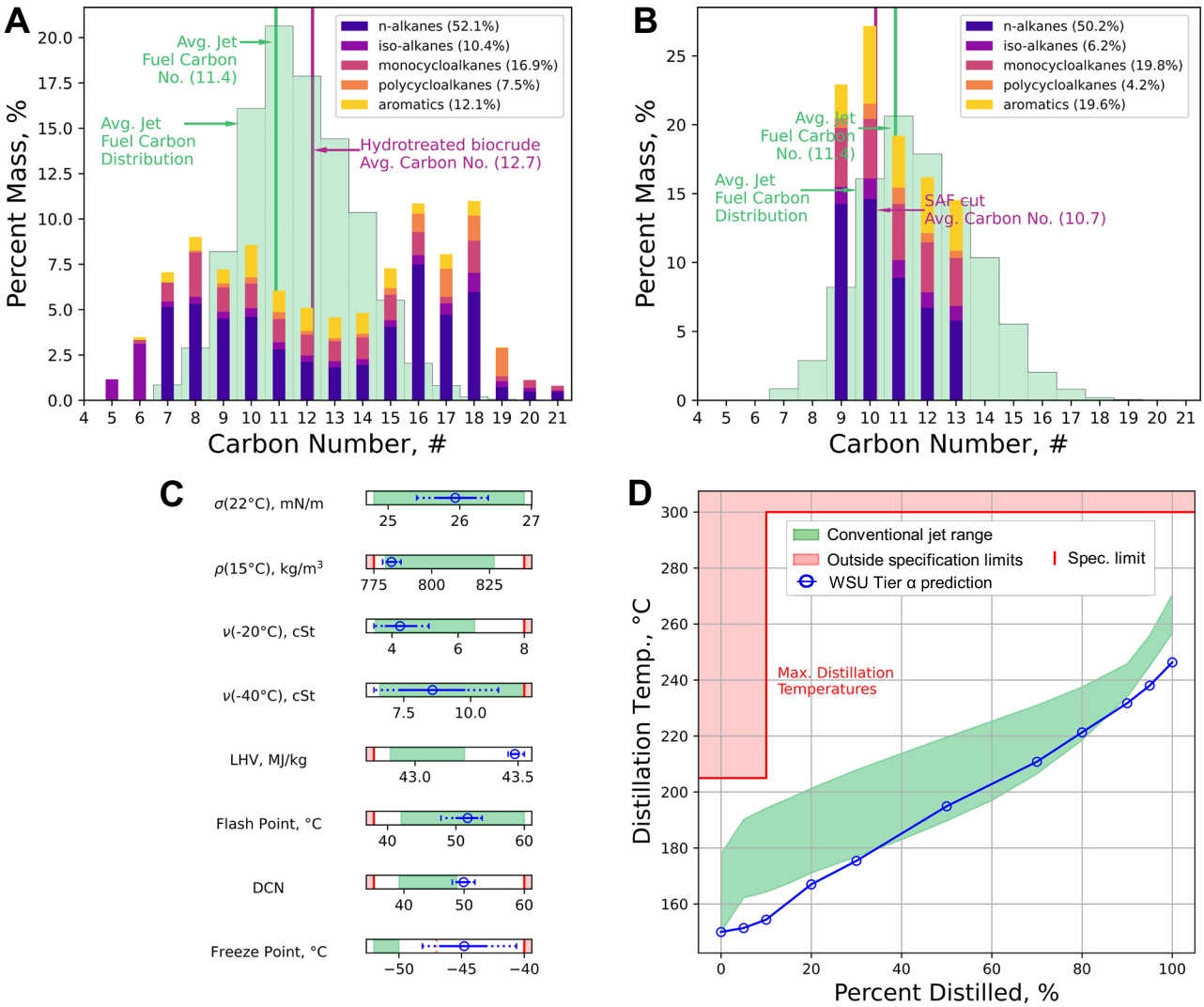


Fig. 4 | ASTM D4054 prescreening of HTL-SAF candidate properties. Tier α results for sustainable aviation fuel (SAF) candidate properties compared to conventional jet fuel, including hydrocarbon type abundance from carbon numbers C5–C21 (n-alkanes, isoalkanes, mono- and poly-cycloalkanes, and aromatics) and average carbon number of **A** GCxGC-FID results for hydrotreated hydrothermal liquefaction biocrude oil and **B** GCxGC-FID results for SAF cut from hydrotreated HTL biocrude oil; **C** predicted jet fuel properties for surface tension, density, viscosity, lower heating value (LHV), flash point, derived cetane number (DCN), and freeze point using 10,000 rounds of random sampling from a hydrocarbon database for bulk properties prediction, where the center represents the determined average, solid error bars represent 68% confidence interval, and dashed error bars represent 95% confidence interval; and **D** predicted ASTM D86 distillation curve. The average conventional jet fuel experience range (green shaded region), specification limit (red vertical line), and outside the specification limit (red shaded region) is also illustrated for reference. Source data are provided as a Source Data file.

It was observed that the surface tension (σ), density (ρ), kinematic viscosity (ν), lower heating value (LHV), flash point, derived cetane number, and freeze point were all within ASTM specification limits (Fig. 4C). Notably, the density and freeze point of the SAF candidate were near, but still, within ASTM specification limits, which was attributed to the high amount of heavy-end n-alkanes. At the same time, LHV was even higher than conventional jet fuel, likely due to the higher concentration of n-alkanes, which have high specific energy[43]. Although n-alkanes have high energy content and thermal stability, light n-alkanes (<nC7) can cause flash point violations and heavy n-alkanes (>nC14) can cause freeze point violations[43]. Since the SAF cut contained >50% n-alkanes, the fuel likely has a high smoke point, preventing soot formation and enabling cleaner combustion. However, there were few isoalkanes present in the hydrotreated HTL biocrude oil and SAF cut. Lightly branched isoalkanes are desirable for jet fuel because they improve cold flow properties, specific energy, thermal stability, and freeze point[43–45]. Isoalkanes improve the pour point and cloud point of jet fuel, which are especially important for the operability and performance of jet fuel at high altitude and low temperature[46,47]. Meanwhile, the high content of cycloalkanes and aromatics (~44%) also help to improve performance and handling properties of the SAF. For instance, monocycloalkanes improve the density, freeze point, and specific energy of the fuel, while polycycloalkanes increase energy density, specific energy, and thermal stability[43]. Aromatics are needed to promote seal swelling in jet engines, but they have high particulate emissions and cause significant sooting during burning. Lastly, the predicted ASTM D86 distillation curve of the SAF candidate was similar to the conventional jet range (Fig. 4D). Although the light and heavy ends fell slightly below the jet range, it was still within ASTM specification limits. The final boiling point for the SAF cut was set lower than the typical range for conventional jet fuel to address concerns about the freezing point. To further mitigate the freezing point issue, an additional isomerization step could be implemented to convert linear alkanes into lightly branched alkanes. This adjustment would allow the final boiling point to be increased to the conventional jet fuel range, resulting in a higher SAF yield. This food waste-derived HTL-SAF candidate was predicted to meet key ASTM specifications without blending. However, it is noted that some parameters, such as density and freeze point, are close to the specification limits. Therefore, blending may be required if any variations in feedstock composition cause deviations in the resulting fuel properties. The mass, carbon, and energy yields for this SAF candidate (Table S7) provided implications for scale-up and future areas of direction. After HTL biocrude pretreatment, hydrotreating, and distillation to obtain the SAF cut, the final mass yield from food waste was 6.94 wt %. For instance, improved mass transfer during hydrotreating could increase the oil yield and decrease losses to char and gaseous products. As mentioned before, increased alkane branching can also help improve the hydrotreated biocrude's final boiling point, resulting in a higher SAF yield. Therefore, continuous hydrotreating, reaction tuning, and catalyst development present potential areas of work to improve viability of the HTL-SAF route.

Circularity index

Sustainability analysis using a circularity index (CI) demonstrated improved circularity in the U.S. aviation industry when implementing a HTL-SAF pathway. CI is a single value, between 0 and 1, that measures the circularity of concern in terms of consumable, renewable, and recoverable resources[48]. In this case, the energy and carbon CI was determined for the entire existing U.S. aviation industry's jet fuel consumption and potential for SAF production from disposed food waste. Average consumption and production data for petroleum products, food waste, and the HTL process was collected for eight value chain categories: extraction, refining, distribution, consumption, disposal, recovery, remake, and reuse (Tables S8 and S9). In the existing U.S. aviation industry, jet fuel is mainly derived from petroleum products, with very little SAF consumption. In the first half of 2024, an average of 1,657,000 barrels of jet fuel were consumed per day[49]. Meanwhile, just 15.8 million gallons of SAF were consumed in 2022[50]. In 2019, it was estimated that 66.2 million wet tons of food waste was managed in the U.S., with 39.6 million tons going to landfills[51].

After implementing the use of HTL and hydrotreating for food waste recoverable from landfills, the CIs improved significantly. Energy CI increased from 1.35 to 32.51% and carbon CI increased from 0.13% to 17.12%. Despite the improved circularity through producing SAF from food waste, it is apparent that food waste alone will only be part of the solution to provide feedstocks for SAF. Specifically, municipal solid waste and wet waste (including food waste) is just a fraction of the potential sustainable biomass supply for production of liquid transportation fuels. Other emerging biomass feedstocks, such as algae, energy crops, forestry, agricultural, vegetable oil, and starch supplies, also have potential for SAF production[52]. Additionally, HTL biocrude from food waste can produce other bioproducts besides SAF (such as gasoline, diesel, and biobinders), so the potential CI of the overall system could be much higher[53]. Thereby, while food waste alone can help increase energy and carbon CI of the aviation industry by nearly 20%, utilization of other biomass feedstocks for SAF production can fill the gap to create a fully circular bioeconomy in the aviation industry and beyond.

The pathway from food waste to SAF via HTL and hydrotreating is complicated, involving many factors and variations. To consider the uncertainty in CI under changes in pathway conditions, including amount of disposed and recoverable food waste, carbon and energy contents of food waste and SAF, and mass yields during hydrotreating and distillation, a Monte Carlo analysis with 1,000 simulations was conducted. A larger range of uncertainty was depicted for energy circularity, ranging from 28.26% to 48.73% compared to 14.05–26.93% for carbon circularity (Fig. 5A). For both CIs, higher circularity was observed at increased food waste recovery, which was reflected in its high Spearman's rank correlation coefficient (Fig. 5B). In the U.S., 59.84% of food waste is managed by landfills, with the other ~40% managed by combustion, donation, sewer/wastewater treatment, composting, biochemical processing, animal feed, anaerobic digestion, and land application[51]. Therefore, the production of SAF and value of CI for the aviation industry could continue to increase if food waste management beyond landfills was targeted for recovery in HTL and biorefinery processes. The carbon and energy content of the food waste also had a relatively high correlation to their respective CIs, indicating the importance of the feedstock's biochemical composition. Overall, increased recovery of food waste from other waste management practices would most greatly improve the circularity of this HTL-SAF pathway.

HTL is a promising pathway to produce SAF. Although HTL can valorize food waste into biocrude oil, hydrotreating is still required to remove disadvantageous heteroatoms. It was determined that temperature, catalyst:oil ratio, and RT were the most impactful parameters on hydrotreating for de-O, carbon recovery, and energy recovery. These parameters were optimized to achieve high de-O and de-N, producing a SAF sample that met ASTM specification limits for several fuel properties without blending. Future work may explore parametric tuning and catalyst development to promote isomerization during hydrotreating for improved cold flow properties and higher SAF yield. Furthermore, this pathway also demonstrated improved energy and carbon circularity in the U.S. aviation sector, which could be further increased by changing food waste management practices for higher food waste recovery. In all, it was shown that HTL of food waste is a potential pathway to produce drop-in SAF for a decarbonized aviation industry.

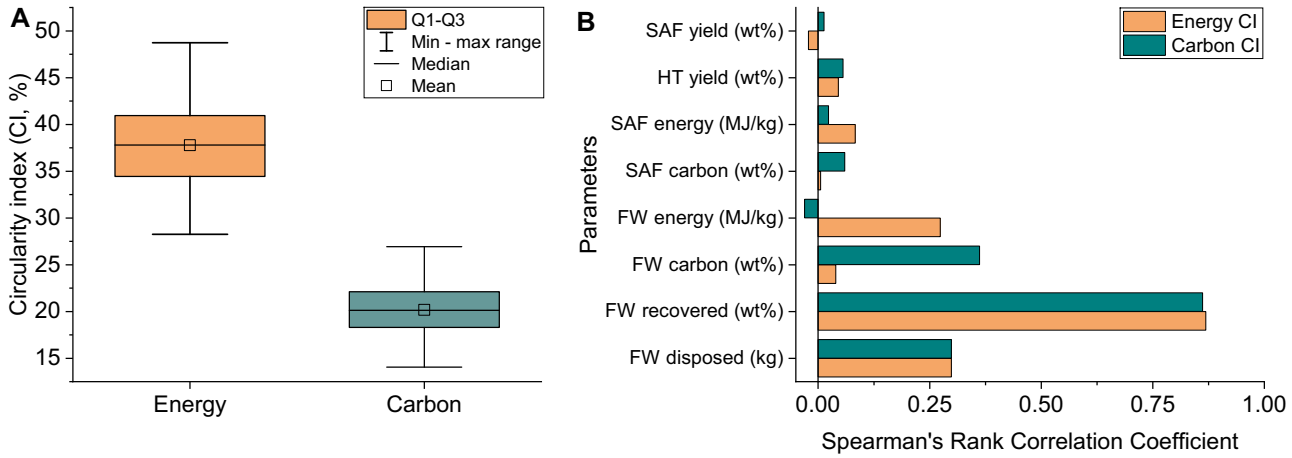


Fig. 5 | Results of the uncertainty and sensitivity analysis from 1,000 trial random sampling in a Monte Carlo simulation. A Box and whisker plot of the uncertainty range for energy and carbon circularity index (CI), where the shaded region represents the lower (Q1) through upper (Q3) quartile, the whiskers represent the minimum and maximum values, the center line represents the median and the center point represents the mean. **B** Spearman's Rank Correlation Coefficient (correlation coefficient of the corresponding parameter and CI value) for the sensitivity of energy and carbon CI for the HTL-SAF pathway, where FW food waste, HT hydrotreating. Source data are provided as a Source Data file.

Methods

HTL biocrude oil

Food waste from a local food processing plant in Champaign, IL was processed in a pilot-scale continuous plug flow HTL reactor at 20 wt% solids content, 300 °C and 12.4 MPa for a 20 min RT[23]. Prior to hydrotreating, multi-stage pretreating was performed to desalt the biocrude using a water wash at 100 °C for 60 min; dewater according to ASTM D2892 Annex X1; and deash through isolation of fraction <350 °C via distillation[39]. The physicochemical characterization of the food processing waste feedstock, initial (BC), and pretreated (PT) HTL biocrude oil are in Table S2.

Hydrotreating experiments

Commercial catalyst $CoMo/Al_2O_3$ (ThermoFisher Scientific) was used to hydrotreat HTL biocrude oil. First, experiments were completed in triplicate for a Taguchi design L9 orthogonal array (Table S3) to screen the significance of temperature, catalyst-to-oil ratio (cat:oil w/w), RT, and hydrogen load-to-biocrude oxygen ratio (H:O mol:mol). Then, experiments were completed in duplicate using a Box-Behnken design to optimize the parameter levels using response surface methodology (Table S4). Designated amounts of biocrude oil and catalyst were loaded into a 100 ml Series 4590 Micro Stirred Reactor (Parr Instrument Company, IL, USA). The reactor was purged to remove air, pressurized with hydrogen, then heated to desired reaction conditions with a stir rate of 350 rpm throughout the RT. After completion of the hydrotreating reaction, the reactor was quenched with water to cool. The oil and aqueous phases were collected in a pre-weighed container and separated by centrifugation. The catalyst and char products were collected separately, rinsed with DCM, and vacuum filtered. The gaseous phase was sampled in a gas bag (Sigma-Aldrich) for analysis. The reactor was rinsed with DCM to recover any residual oil, char, or catalyst. The DCM evaporated overnight at 60 °C and the final oil and char mass was recorded. The oil recovered from the solvent rinse was not analyzed. A commercial Jet-A sample was obtained from Willard Airport at the University of Illinois Urbana-Champaign (Savoy, IL, USA).

The mass yields, carbon recovery, and energy recovery were calculated according to Eqs. 1–3, respectively.

$$m_x = \frac{m_x}{m_b} \times 100\% \tag{1}$$

$$C_r = \frac{C_o}{C_b} \times m_o \times 100\% \tag{2}$$

$$E_r = \frac{HHV_o}{HHV_b} \times m_o \times 100\% \tag{3}$$

where m = mass of product, x = product type (o = oil, w = water, g = gas, c = char, b = biocrude), C = carbon content (r = recovered), E_r = energy recovery, and HHV = higher heating value. Oil and aqueous products were directly measured, while char was calculated by subtracting catalyst mass from the collected solids product, and gas was calculated by difference.

Heteroatom removal (de-X) was calculated based on the elemental composition of the HTL biocrude (Eq. 4):

$$de - X = 1 - \frac{X_{\text{final}}}{X_{\text{initial}}} \tag{4}$$

where X = heteroatom element (ex. oxygen (O), nitrogen (N), or sulfur (S)), X_{final} = content in the upgraded biocrude, and X_{initial} = content in the initial HTL biocrude.

Analytical methods

CE440 CHN analyzer (Exeter Analytical, Coventry, UK) was used to determine the elemental composition (carbon, hydrogen, and nitrogen content). Triplicate measurements were taken from each sample, where tins were cold-weld shut prior to purging to minimize loss of volatiles. Due to the presence of air in the sample, nitrogen content was recalculated based on a blank. The detection limit was 0.10% for all elements. Oxygen was determined by difference. A PerkinElmer ICP-MS (Model NexION 350D) was used to measure sulfur content. Due to sample volume limitations, this was performed once per sample. Dulong's formula[54] (Eq. 5) was used to calculate the higher heating value (HHV).

$$HHV\ (\text{MJ/kg}) = 0.3516\,C + 1.16225\,H - 0.1109\,O + 0.0628\,N \tag{5}$$

The weight loss curves and boiling point distribution were determined with a Q50 thermogravimetric analyzer (TGA) (TA Instruments, DE, USA). 15 mg of sample was heated from 20 to 600 °C at 20 °C/min and

nitrogen flow rate of 60 mL/min. The temperatures for boiling point ranges were selected based on literature[55]. It is important to note that thermal degradation may occur due to atmospheric conditions during TGA, making boiling points less accurate. The total acid number (TAN) was determined according to ASTM D974, in which oil sample dissolved in a solution of toluene, 2-propanol, and water was titrated with phenolphthalein indicator and 0.1 M KOH[56]. Chemical composition was determined via gas chromatography-mass spectrometry (GC-MS), using a Micromass 70-VSE (Waters Corporation, MA, USA). 2 µL sample injected split mode in a ZB-5MS column. 30–80 °C at 25 °C/min and 2 min hold, then increased to 220 °C at 8 °C/min, then to 280 °C at 25 °C/min. A 70 eV electron ionization voltage was used, scanning from 30 to 800 m/z. Peaks were analyzed using MassLynx Mass Spectrometry Software according to the NIST08 Mass Spectral Database[57].

The jet fuel cut (150–230 °C) of hydrotreated HTL biocrude oil from this work was characterized using Tier α tests as a precursor to ASTM D4054 evaluation. Viability of the SAF candidate was prescreened by comparing its chemical composition, distillation, and fuel properties to conventional jet fuel and ASTM D7566 specifications. Specifically, a 2D-GC method developed by the Heyne group at Washington State University was used to determine the hydrocarbon type and distribution, a distillation curve was obtained from the 2D-GC data to show ASTM D86 results, and the 2D-GC results were used to predict jet fuel properties[37,38,42].

The circularity index (CI, Eq. 6) of the system was calculated according to methods previously described[48]:

$$CI = \frac{Y_R}{Y} = \frac{\sum_{i=1}^{n} R_i X_i + \left[a_{ij}\right]\left[X_i\right]}{\sum_{i=1}^{n} (C_i + R_i) X_i} \quad (6)$$

where Y_R is the total renewable and recovered resources in a circular system, Y is the total output of the system, R_i is the fraction from renewable resources, X_i is the content of the ith category, a_{ij} is the fraction from recoverable resources, and C_i is the fraction from consumable resources. Uncertainty and sensitivity analysis of the CI was done using Monte Carlo analysis with 1000 simulations. The Spearman's rank correlation coefficient was used to rank the correlation of each parameter between the CIs. Further explanations of the CI method, uncertainty, and sensitivity methods are in the Supplementary Information.

Data analysis was conducted in Excel (Version 2507) and OriginPro 2025 (Version 10.2 SR1).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data are provided with this paper: The data that supports the findings of this study are provided in the Source Data file. Source data are provided with this paper.

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Author contributions

S.S. and S.Y[1]. performed the biocrude oil hydrotreating experiments. S.S. performed the physicochemical characterization of biocrude oil, calculated the CI values, and wrote the manuscript. B.S., Z.W., J.W., S.Y[3]. and Z.Y. participated in discussion of the fuel property analysis and design of experiments. Z.Y. and J.S.H. performed the SAF Tier α property prescreening. H.K. produced the HTL biocrude oil used in this project. Y.Z. supervised and directed this project. All authors commented on the manuscript.

Competing interests
The authors declare no competing interests.

Additional information
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